

EXAMPLE 3.1

Heating Ammonia at Constant Pressure

A vertical piston–cylinder assembly containing 0.1 lb of ammonia, initially a saturated vapor, is placed on a hot plate. Due to the weight of the piston and the surrounding atmospheric pressure, the pressure of the ammonia is 20 lbf/in.^2 . Heating occurs slowly, and the ammonia expands at constant pressure until the final temperature is 77°F . Show the initial and final states on T – v and p – v diagrams, and determine

- (a) the volume occupied by the ammonia at each end state, in ft^3 .
- (b) the work for the process, in Btu.

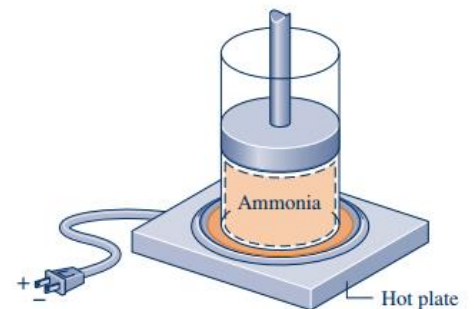
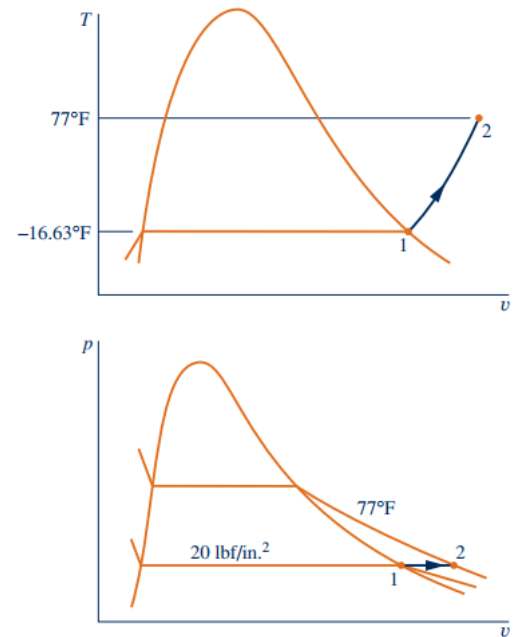


TABLE A-15E

Properties of Superheated Ammonia Vapor

TABLE A-15E

(Continued)

T °F	v ft ³ /lb	u Btu/lb	h Btu/lb	s Btu/lb · °R	v ft ³ /lb	u Btu/lb	h Btu/lb	s Btu/lb · °R
$p = 18 \text{ lbf/in.}^2$ ($T_{\text{sat}} = -20.60^\circ\text{F}$)					$p = 20 \text{ lbf/in.}^2$ ($T_{\text{sat}} = -16.63^\circ\text{F}$)			
Sat.	14.896	554.75	604.40	1.3775	13.497	555.78	605.76	1.3687
-20	14.919	555.00	604.72	1.3783				
-10	15.306	559.13	610.14	1.3905	13.730	558.55	609.40	1.3769
0	15.688	563.23	615.52	1.4023	14.078	562.70	614.84	1.3888
10	16.068	567.31	620.87	1.4138	14.422	566.83	620.24	1.4005
20	16.444	571.37	626.18	1.4250	14.764	570.94	625.61	1.4118
30	16.818	575.42	631.47	1.4359	15.103	575.02	630.95	1.4228
40	17.189	579.46	636.75	1.4466	15.439	579.09	636.26	1.4335
50	17.558	583.48	642.00	1.4570	15.773	583.14	641.55	1.4440
60	17.925	587.50	647.25	1.4672	16.105	587.19	646.83	1.4543
70	18.291	591.52	652.48	1.4772	16.436	591.23	652.10	1.4643
80	18.655	595.53	657.71	1.4869	16.765	595.26	657.35	1.4741
90	19.018	599.55	662.94	1.4965	17.094	599.30	662.60	1.4838
$p = 30 \text{ lbf/in.}^2$ ($T_{\text{sat}} = -0.57^\circ\text{F}$)					$p = 40 \text{ lbf/in.}^2$ ($T_{\text{sat}} = 11.65^\circ\text{F}$)			
Sat.	9.2286	559.77	611.04	1.3349	7.0414	562.60	614.76	1.3109
0	9.2425	560.02	611.36	1.3356				
10	9.4834	564.38	617.07	1.3479				
20	9.7209	568.70	622.70	1.3598	7.1965	566.39	619.69	1.3213
30	9.9554	572.97	628.28	1.3713	7.3795	570.86	625.52	1.3333
40	10.187	577.21	633.80	1.3824	7.5597	575.28	631.28	1.3450
50	10.417	581.42	639.28	1.3933	7.7376	579.65	636.96	1.3562
60	10.645	585.60	644.73	1.4039	7.9134	583.97	642.58	1.3672
70	10.871	589.76	650.15	1.4142	8.0874	588.26	648.16	1.3778
80	11.096	593.90	655.54	1.4243	8.2598	592.52	653.69	1.3881
90	11.319	598.04	660.91	1.4342	8.4308	596.75	659.20	1.3982
100	11.541	602.16	666.27	1.4438	8.6006	600.97	664.67	1.4081
110	11.762	606.28	671.62	1.4533	8.7694	605.17	670.12	1.4178
$p = 50 \text{ lbf/in.}^2$ ($T_{\text{sat}} = 21.65^\circ\text{F}$)					$p = 60 \text{ lbf/in.}^2$ ($T_{\text{sat}} = 30.19^\circ\text{F}$)			
Sat.	5.7049	564.78	617.60	1.2923	4.8009	566.53	619.87	1.2770
40	5.9815	573.30	628.68	1.3149	4.9278	571.25	626.00	1.2894
60	6.2733	582.31	640.39	1.3379	5.1788	580.60	638.14	1.3133
80	6.5574	591.10	651.82	1.3595	5.4218	589.66	649.90	1.3355
100	6.8358	599.75	663.04	1.3799	5.6587	598.52	661.39	1.3564
120	7.1097	608.30	674.13	1.3993	5.8910	607.23	672.68	1.3762
140	7.3802	616.80	685.13	1.4180	6.1198	615.86	683.85	1.3951
160	7.6480	625.28	696.09	1.4360	6.3458	624.44	694.95	1.4133
200	8.1776	642.27	717.99	1.4702	6.7916	641.59	717.05	1.4479
240	8.7016	659.44	740.00	1.5026	7.2318	658.87	739.21	1.4805
280	9.2218	676.88	762.26	1.5336	7.6679	676.38	761.58	1.5116
320	9.7391	694.65	784.82	1.5633	8.1013	694.21	784.22	1.5414
360	10.254	712.79	807.73	1.5919	8.5325	712.40	807.20	1.5702